

Report for ForestQuery into Global Deforestation, 1990 to 2016

ForestQuery is on a mission to combat deforestation around the world and to raise awareness about this topic and its impact on the environment. The data analysis team at ForestQuery has obtained data from the World Bank that includes forest area and total land area by country and year from 1990 to 2016, as well as a table of countries and the regions to which they belong.

The data analysis team has used SQL to bring these tables together and to query them in an effort to find areas of concern as well as areas that present an opportunity to learn from successes.

1. GLOBAL SITUATION

According to the World Bank, the total forest area of the world was 41282694.9 sq km in 1990. As of 2016, the most recent year for which data was available, that number had fallen to 39958245.9 sq km, a loss of 1324449 sq km or 3.20%.

The forest area lost over this time period is slightly more than the entire land area of Peru listed for the year 2016 (which is 1279999.9 sq km).

2. REGIONAL OUTLOOK

In 2016, the percent of the total land area of the world designated as forest was 31.4%. The region with the highest relative forestation was Latin America & Caribbean, with 46.2%, and the region with the lowest relative forestation was Middle East & North Africa, with 2% forestation.

In 1990, the percent of the total land area of the world designated as forest was 32.4%. The region with the highest relative forestation was Latin America & Caribbean, with 51%, and the region with the lowest relative forestation was Middle East & North Africa, with 1.8% forestation.

Table 2.1: Percent Forest Area by Region, 1990 & 2016:

Region	1990 Forest Percentage	2016 Forest Percentage
Latin America & Caribbean	51%	46.2%
Europe & Central Asia	37.3%	38%
North America	35.7%	36%
World	32.4%	31.4%
Sub-Saharan Africa	30.7%	28.8%
East Asia & Pacific	25.8%	26.4%
South Asia	16.5%	17.5%
Middle East & North Africa	1.8%	2%

The only regions of the world that decreased in percent forest area from 1990 to 2016 were Latin America & Caribbean (dropped from 51% to 46.2%) and Sub-Saharan Africa (30.7% to 28.8%). All other regions actually increased in forest area over this time period. However, the drop in forest area in the two aforementioned regions was so large, the percent forest area of the world decreased over this time period from 32.4% to 31.4%.

3. COUNTRY-LEVEL DETAIL

A. SUCCESS STORIES

There is one particularly bright spot in the data at the country level, China. This country actually increased in forest area from 1990 to 2016 by 527229 sq km. It would be interesting to study what has changed in this country over this time to drive this figure in the data higher. The country with the next largest increase in forest area from 1990 to 2016 was the United States, but it only saw an increase of 79200 sq km, much lower than the figure for China.

China and United States are of course very large countries in total land area, so when we look at the largest *percent* change in forest area from 1990 to 2016, we aren't surprised to find a much smaller country listed at the top. Iceland increased in forest area by 213.7% from 1990 to 2016.

B. LARGEST CONCERNS

Which countries are seeing deforestation to the largest degree? We can answer this question in two ways. First, we can look at the absolute square kilometer decrease in forest area from 1990

to 2016. The following 3 countries had the largest decrease in forest area over the time period under consideration:

Table 3.1: Top 5 Amount Decrease in Forest Area by Country, 1990 & 2016:

Country	Region	Absolute Forest Area Change
Brazil	Latin America & Caribbean	541510
Indonesia	East Asia & Pacific	282194
Myanmar	East Asia & Pacific	107234
Nigeria	Sub-Saharan Africa	106506
Tanzania	Sub-Saharan Africa	102320

The second way to consider which countries are of concern is to analyze the data by percent decrease.

Table 3.2: Top 5 Percent Decrease in Forest Area by Country, 1990 & 2016:

Country	Region	Pct Forest Area Change
Togo	Sub-Saharan Africa	75%
Nigeria	Sub-Saharan Africa	61.8%
Uganda	Sub-Saharan Africa	59%
Mauritania	Sub-Saharan Africa	46.7%
Honduras	Latin America & Caribbean	45%

When we consider countries that decreased in forest area percentage the most between 1990 and 2016, we find that four of the top 5 countries on the list are in the region of Sub-Saharan Africa. The countries are Togo, Nigeria, Uganda and Mauritania. The 5th country on the list is Honduras, which is in the Latin America & Caribbean region.

From the above analysis, we see that Nigeria is the only country that ranks in the top 5 both in terms of absolute square kilometer decrease in forest as well as percent decrease in forest area from 1990 to 2016. Therefore, this country has a significant opportunity ahead to stop the decline and hopefully spearhead remedial efforts.

C. QUARTILES

Table 3.3: Count of Countries Grouped by Forestation Percent Quartiles, 2016:

Quartile	Number of Countries
Quartile 1 (0-25%)	85
Quartile 2 (25-50%)	72
Quartile 3 (50-75%)	38
Quartile 4 (75-100%)	9

The largest number of countries in 2016 were found in the 1st quartile.

There were 9 countries in the top quartile in 2016. These are countries with a very high percentage of their land area designated as forest. The following is a list of countries and their respective forest land, denoted as a percentage.

Table 3.4: Top Quartile Countries, 2016:

Country	Region	Pct Designated as Forest
Suriname	Latin America & Caribbean	98.2%
Micronesia, Fed. Sts.	East Asia & Pacific	91.8%
Gabon	Sub-Saharan Africa	90%
Seychelles	Sub-Saharan Africa	88.4%
Palau	East Asia & Pacific	87.6%
American Samoa	East Asia & Pacific	87.5%
Guyana	Latin America & Caribbean	84%
Lao PDR	East Asia & Pacific	82.1%
Solomon Islands	East Asia & Pacific	77.9%

In 2016, the United States had 33.9% of its land area designated as forest. Comparing this against all other countries, 94 countries had a higher percentage of forest area than the United States. This indicates that while the U.S. saw a net increase in absolute forest area between 1990 and 2016, it still ranks below nearly half of the world's countries in relative forestation, reinforcing that total land area and forest percentage tell different stories about a country's environmental footprint.

4. RECOMMENDATIONS

Based on the World Bank dataset, we can observe a large global forestation loss combined with successful localized conservation.

The scale of net forest loss is very significant, with approximately 1.3 million square kilometers of forest lost between 1990 and 2016 which is roughly equivalent to the size of Peru.

Interestingly the data also shows how deforestation is not a uniform global process, regions like Europe & Central Asia and East Asia & Pacific saw a net increase in forest area due to aggressive reforestation policies where Latin America & Caribbean and Sub-Saharan Africa saw the most drastic declines, Latin America specifically dropped from 51% forestation in 1990 to around 46% in 2016. And while some countries like China and the US gained forest area, these gains were not enough to offset the massive losses in tropical regions.

To make an impact, we must learn from successful growth stories and focus on countries where significant forest areas were lost. Those countries represent the front line in of deforestation, Brazil by far is the largest absolute forest loss globally, as home to majority of the amazon it affects global climate stability more than any other region. We can learn from countries like China for taking the global lead in reforestation through massive governments programs like the "Great Green Wall". China proved that adding hundreds of thousands of square kilometers of forest is possible.

5. APPENDIX: SQL Queries Used

PART 1:

```
CREATE VIEW forestation AS
```

```
SELECT f.*, l.total_area_sq_mi, r.region, r.income_group, (f.forest_area_sqkm  
/(l.total_area_sq_mi * 2.59) ) * 100 as percent_land
```

```
FROM regions r
```

```
JOIN forest_area f
```

```
ON r.country_code=f.country_code
```

```
JOIN land_area l ON l.country_code=f.country_code AND f.year=l.year ;
```

```
SELECT country_name, forest_area_sqkm  
  
FROM forestation  
  
WHERE year = 1990 and country_name = 'World' ;
```

```
SELECT country_name, forest_area_sqkm  
  
FROM forestation  
  
WHERE year = 2016 and country_name = 'World' ;
```

```
WITH data_1990 AS (  
  
    SELECT country_name, forest_area_sqkm  
  
    FROM forestation  
  
    WHERE year = 1990  
  
) ,  
data_2016 AS (  
  
    SELECT country_name, forest_area_sqkm  
  
    FROM forestation  
  
    WHERE year = 2016  
  
)  
SELECT  
  
    f1.country_name,  
  
    f1.forest_area_sqkm AS forest_1990,  
  
    f2.forest_area_sqkm AS forest_2016,  
  
    (f2.forest_area_sqkm - f1.forest_area_sqkm) AS absolute_change_sqkm,
```

```
((f2.forest_area_sqkm - f1.forest_area_sqkm) / f1.forest_area_sqkm) * 100 AS  
percent_change
```

```
FROM data_1990 f1
```

```
JOIN data_2016 f2 ON f1.country_name = f2.country_name
```

```
ORDER BY absolute_change_sqkm ASC;
```

```
SELECT total_area_sq_mi*2.59 AS area, country_name
```

```
from forestation
```

```
where total_area_sq_mi*2.59 between 1000000 and 1400000
```

```
order by total_area_sq_mi*2.59 desc;
```

PART 2:

```
WITH data_1990 AS (
```

```
    SELECT region, forest_area_sqkm, total_area_sq_mi
```

```
    FROM forestation
```

```
    WHERE year = 1990
```

```
),
```

```
data_2016 AS (
```

```
    SELECT region, forest_area_sqkm, total_area_sq_mi
```

```
    FROM forestation
```

```
    WHERE year = 2016
```

```
),
```

```
region_totals AS (
```

```
    SELECT
```

```

    f1.region,

    SUM(f1.forest_area_sqkm) AS total_forest_1990,

    SUM(f2.forest_area_sqkm) AS total_forest_2016,

    SUM(f1.total_area_sq_mi * 2.59) AS total_land_1990,

    SUM(f2.total_area_sq_mi * 2.59) AS total_land_2016

FROM data_1990 f1

JOIN data_2016 f2 ON f1.region = f2.region

GROUP BY f1.region

)

SELECT

    region,

    (total_forest_1990 / total_land_1990) * 100 AS percent_forest_1990,

    (total_forest_2016 / total_land_2016) * 100 AS percent_forest_2016

FROM region_totals

ORDER BY percent_forest_2016 DESC;

```

PART 3:

A)

```

WITH data_1990 AS (

    SELECT country_name, region, forest_area_sqkm AS forest_1990

    FROM forestation

    WHERE year = 1990

),

data_2016 AS (

    SELECT country_name, region, forest_area_sqkm AS forest_2016

```

```
FROM forestation

WHERE year = 2016

)

SELECT

    f1.country_name,

    f1.region,

    (f2.forest_2016 - f1.forest_1990 ) AS area_increase

FROM data_1990 f1

JOIN data_2016 f2 ON f1.country_name= f2.country_name

WHERE f1.country_name != 'World'

AND forest_1990 IS NOT NULL

AND forest_2016 IS NOT NULL

ORDER BY area_increase DESC

LIMIT 5;


WITH data_1990 AS (

    SELECT country_name, region, forest_area_sqkm AS forest_1990

    FROM forestation

    WHERE year = 1990

),

data_2016 AS (

    SELECT country_name,region,forest_area_sqkm AS forest_2016

    FROM forestation

    WHERE year = 2016
```

)

SELECT

f1.country_name,

f1.region,

((f2.forest_2016 - f1.forest_1990) / f1.forest_1990) * 100 AS percent_increase

FROM data_1990 f1

JOIN data_2016 f2 ON f1.country_name= f2.country_name

WHERE f1.country_name != 'World'

AND forest_1990 IS NOT NULL

AND forest_2016 IS NOT NULL

ORDER BY percent_increase DESC

LIMIT 5;

B)

WITH data_1990 AS (

SELECT country_name, region, forest_area_sqkm AS forest_1990

FROM forestation

WHERE year = 1990

),

data_2016 AS (

SELECT country_name, region, forest_area_sqkm AS forest_2016

FROM forestation

WHERE year = 2016

)

SELECT

f1.country_name,

f1.region,

(f1.forest_1990 - f2.forest_2016) AS area_decrease

FROM data_1990 f1

JOIN data_2016 f2 ON f1.country_name= f2.country_name

WHERE f1.country_name != 'World'

AND forest_1990 IS NOT NULL

AND forest_2016 IS NOT NULL

```
ORDER BY area_decrease DESC
LIMIT 5;
```

```
WITH data_1990 AS (
    SELECT country_name, region, forest_area_sqkm AS forest_1990
    FROM forestation
    WHERE year = 1990
),
data_2016 AS (
    SELECT country_name, region, forest_area_sqkm AS forest_2016
    FROM forestation
    WHERE year = 2016
)
SELECT
    f1.country_name,
    f1.region,
    ((f1.forest_1990 - f2.forest_2016) / f1.forest_1990) * 100 AS percent_decrease
FROM data_1990 f1
JOIN data_2016 f2 ON f1.country_name = f2.country_name
WHERE f1.country_name != 'World'
AND forest_1990 IS NOT NULL
AND forest_2016 IS NOT NULL
ORDER BY percent_decrease DESC
LIMIT 5;
```

c)

```
WITH data_2016 AS (
    SELECT country_name,
        (SUM(forest_area_sqkm) / NULLIF(SUM(total_area_sq_mi * 2.59), 0)) * 100 AS
        forest_percentage
    FROM forestation
    WHERE year = 2016
    AND country_name != 'World'
    GROUP BY country_name
)
```

```

SELECT
    CASE
        WHEN forest_percentage <= 25 THEN 'Quartile 1 (0-25%)'
        WHEN forest_percentage <= 50 THEN 'Quartile 2 (25-50%)'
        WHEN forest_percentage <= 75 THEN 'Quartile 3 (50-75%)'
        ELSE 'Quartile 4 (75-100%)' END AS forest_quartile,
    COUNT(*) AS country_count
FROM data_2016
WHERE forest_percentage IS NOT NULL
GROUP BY 1
ORDER BY 1;

```

```

WITH data_2016 AS (
    SELECT country_name,
           region,
           (forest_area_sqkm / NULLIF(total_area_sq_mi * 2.59, 0)) * 100 AS forest_percentage
    FROM forestation
    WHERE year = 2016
)
SELECT

```

country_name, region, forest_percentage

FROM data_2016

WHERE forest_percentage > 75

ORDER BY forest_percentage DESC;

WITH us_2016 AS (

SELECT (forest_area_sqkm / NULLIF(total_area_sq_mi * 2.59, 0)) * 100 AS us_pct

FROM forestation

WHERE year = 2016 AND country_name = 'United States'

),

data_2016 AS (

SELECT country_name,

(forest_area_sqkm / NULLIF(total_area_sq_mi * 2.59, 0)) * 100 AS
forest_percentage

FROM forestation

WHERE year = 2016 AND country_name != 'World'

)

SELECT COUNT(*) AS countries_above_us

FROM data_2016, us_2016

WHERE data_2016.forest_percentage > us_2016.us_pct;